



Semester 2 Examinations, 2016/2017

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Module Numerical Analysis II

Paper No 1

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Instructions: **Answer THREE questions.** All questions carry equal marks.

Duration 2 Hours
Number of Pages 3 (including this page)
Discipline Mathematics

Requirements:

Release in Exam Venue	Yes	☒	No	☐
Release to Library	Yes	☒	No	☐
Statistical/ Log Tables	Yes	☒	No	☐
Graph Paper	None			
Log Graph Paper	None			
Graphic material in colour	Yes	☐	No	☒

Q1. Let p_n be a polynomial of degree n that interpolates a function f at the $n + 1$ distinct points: $\{x_0, x_1, \dots, x_n\}$.

- (a) [8 MARKS] Show that such a polynomial exists.
- (b) [14 MARKS] State and prove Cauchy's theorem for $f(x) - p_n(x)$.
- (c) [3 MARKS] Can this theorem be used to conclude that $p_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$? Explain your answer.
- (d) [8 MARKS] Let p_2 be the polynomial that interpolates $f(x) = x \exp(-x)$ at the points $\{-1, 0, 1\}$. Use Cauchy's Theorem to give an upper bound for the error, $|f(x) - p_n(x)|$, at $x = 1/2$. (You don't have to give a formula for p_2).

Q2. (a) [8 MARKS] Define the *natural piecewise cubic (spline) interpolant* to the function f at the $n + 1$ distinct points: $\{x_0, x_1, \dots, x_n\}$.

- (b) [14 MARKS] Suppose that the interpolation points are equally spaced and $h = (x_n - x_0)/n$. Denote by S the cubic spline interpolant to f , and s_i its restriction to $[x_{i-1}, x_i]$. If $\sigma_i := S''(x_i)$, show that

$$s_i(x) = \alpha_i(x - x_{i-1}) + \beta_i(x_i - x) + \frac{\sigma_{i-1}}{6h}(x_i - x)^3 + \frac{\sigma_i}{6h}(x - x_{i-1})^3, \quad (1a)$$

where

$$\alpha_i = \frac{f(x_i)}{h} - \frac{h}{6}\sigma_i, \quad \beta_i = \frac{f(x_{i-1})}{h} - \frac{h}{6}\sigma_{i-1}. \quad (1b)$$

Hence deduce that S can be determined from solving the system: $\sigma_1 = 0$, $\sigma_n = 0$ and

$$\frac{1}{6}(\sigma_{i-1} + 4\sigma_i + \sigma_{i+1}) = \frac{1}{h^2}(f(x_{i-1}) - 2f(x_i) + f(x_{i+1})) \quad \text{for } i = 2, \dots, n-1.$$

- (c) [11 MARKS] Suppose that S is a natural cubic spline on $[0, 2]$ with

$$S(x) = \begin{cases} 3x + a(1-x)^3 + bx^3, & \text{for } 0 \leq x < 1, \\ c(2-x) - (2-x)^3 + d(x-1)^3, & \text{for } 1 \leq x \leq 2. \end{cases}$$

Find a , b , c , and d .

- Q3. (a) [11 MARKS] What is meant by the *precision* of a quadrature method?
Determine the maximum precision of the following quadrature rule for estimating $\int_0^1 f(x)dx$:

$$Q(f) = \frac{1}{3} \left(2f\left(\frac{1}{4}\right) - f\left(\frac{1}{2}\right) + 2f\left(\frac{3}{4}\right) \right).$$

- (b) [11 MARKS]

- (i) Explain *Gaussian* Quadrature.
(ii) Derive the two-point Gaussian Quadrature rule of approximating $\int_{-1}^1 f(x)dx$.
(c) [11 MARKS] Let $\{\hat{p}_n(x)\}_{n=0}^\infty$ be the sequence of orthogonal monic polynomials with respect to the inner product

$$(f, g) := \int_a^b f(x)g(x)dx.$$

- (i) Prove that $\hat{p}_n(x)$ is orthogonal to *all* polynomials of degree less than n .
(ii) Prove that the zeros of the $\hat{p}_n(x)$ are simple (i.e., distinct).
(iii) Prove that the zeros of the $\hat{p}_n(x)$ are all in the interval $[a, b]$.

- Q4. Consider the following differential equation:

$$-u''(x) = 1 \quad \text{on} \quad (0, 3) \quad \text{and} \quad u(0) = u(3) = 0. \quad (2)$$

- (a) [10 MARKS] State the variational formulation of the differential equation in (2).
Show that the solution to the variational problem is unique.
(b) Suppose one wanted to compute an approximation to (2) using the finite element method (FEM) with the usual piecewise linear Galerkin basis (“hat”) functions on the uniform mesh $\{x_0, x_1, \dots, x_{n-1}, x_n\}$.
(i) [4 MARKS] Give a formula for the basis functions. How is the FEM solution expressed in terms of this basis?
(ii) [4 MARKS] If the linear system for the FEM is written as the matrix-vector equation $Ax = b$, give an expression for the entries of A and b in terms of the basis. (You do not have to derive an explicit formula for these entries).
(iii) [4 MARKS] Explain why A is symmetric and tridiagonal.
(c) [11 MARKS] State and prove C  a’s lemma.